Opts 317. 10/5.

R_{ij}^k - construction (Automaton-to-languages)

Last time:

$$R_{ij}^k = R_{ik}^{k-1} (R_{kk}^{k-1})^* R_{kj}^{k-1} + R_{ij}^{k-1}.$$

$$R_{ij}^0 = \begin{cases} a_1 + \dots + a_\ell & \text{if } i \xrightarrow{a_1 \dots a_\ell} j \\ \emptyset & \text{o.w.} \end{cases} \qquad i \neq j$$

$$R_{ii}^0 = \begin{cases} a_1 + \dots + a_\ell + \Lambda & \text{if } i \xrightarrow{a_1 \dots a_\ell} i \\ \Lambda & \text{o.w.} \end{cases}$$

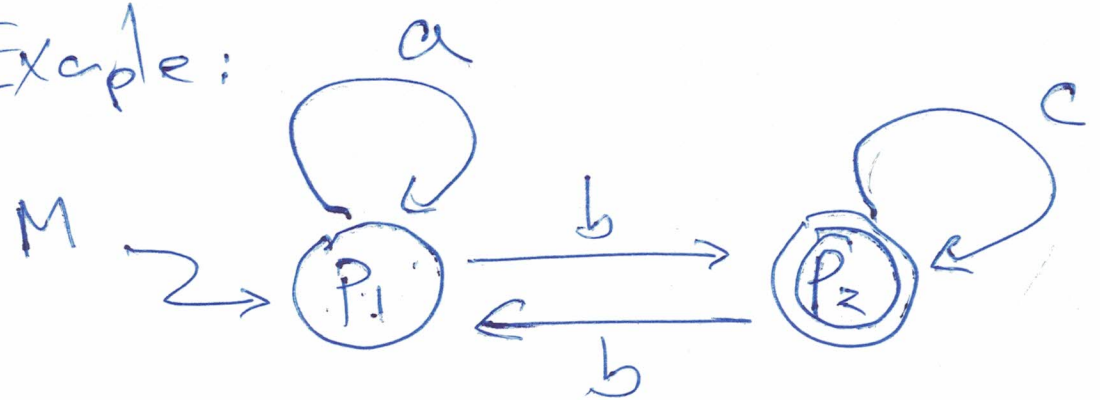
Then $L(M) = R_{1n}^{n}$

# of states.

initial

accepting

Exaple:



$$L(M) = R_{12}^{2}$$

$$= R_{12}^{1}(R_{22}^{1})^{*} R_{22}^{1} + R_{12}^{1}$$

where, $R_{12}^{1} = R_{11}^{0}(R_{11}^{0})^{*} R_{12}^{0} + R_{12}^{0}$

$$R_{22}^{1} = R_{21}^{0}(R_{11}^{0})^{*} R_{12}^{0} + R_{22}^{0}$$

where.

$$R_{11}^{0} = a + \Lambda$$

$$R_{12}^{0} = b.$$

$$R_{21}^{0} = b$$

$$R_{22}^{0} = c + \Lambda$$

So, $R_{12}^{1} = (a+\Lambda)(a+\Lambda)^{*} b + b$

$$R_{22}^{1} = b(a+\Lambda)^{*} b + (c+\Lambda)$$

So, $L(M) = R_{12}^{2} =$

$$((a+\Lambda)(a+\Lambda)^{*} b + b)(b(a+\Lambda)^{*} b + (c+\Lambda))^{*}$$

$$(b(a+\Lambda)^{*} b + (c+\Lambda)) + ((a+\Lambda)(a+\Lambda)^{*} b + \Lambda)$$

Remark:

① . R_{ij}^{k} - construction runs in Exp - time!
the resulting regExp from the construction is of exp - length;

② . The language - to - automaton construction will come up with a $\underbrace{\text{poly-size}}_{\text{small.}}$ NFA.

$\Rightarrow$ ① is harder than ②

$\Rightarrow$ ① is code - understanding;

$\Rightarrow$ ② is code - generation.

Pumping lemma.

logic: try to understand:

$\forall$ dog $\exists$ cat (dog likes cat).

$\forall$ dog $\exists$ cat $\forall$ fish (dog likes cat but cat don't like fish)

$\forall$ dog $\exists$ cat $\forall$ fish $\exists$ snake (- - - - - - - -).

Pumping Lemma ← states a property on all reg. languages.

Let L be an inf. reg language. Then, there is a number $n > 0$ s.t. for any word z in L with $|z| \geq n$, there is a way to write the z into $z = uvw$

such that, $|uv| \leq n$, $|v| \geq 1$, and, for all $i \geq 0$, we have

$$uv^iw \in L.$$

We only use the lemma to prove "a language is not reg.".

Example. Show that $L = \{0^n 1^n : n > 0\}$ is not regular.

Proof. (use PL) Assume that L is reg. From PL, Let n be the number in the lemma.